



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Continuous Assessment Test-II, Winter 2019-2020

CSE2003- Data Structures and Algorithms

Slot: B1

Exam Duration: 90 Minutes

Maximum Marks: 50

Answer all questions (5*10 = 50)

1. Create a structure to store customer id and occupation in a circular single linked list. Engrave pseudo code for insertend (int position, int value), insertinbetween (int position, int value) and deletenode (int n). Also show the processes diagrammatically.
2. Define basic property of a Binary Search Tree (BST) and construct a BST for the given sequence of numbers 45, 36, 76, 23, 89, 115, 98, 39, 41, 56, 69, 48 (Show each insertion trace). Inscribe pseudo code for search (int value) and mention conditions to delete a node from BST.
3. Show how many collisions will occur while mapping given 11 keys with the hash function $h(x) = x \bmod m$. Apply the following methods on the following keys to resolve the collisions.
Keys: 12, 44, 13, 88, 23, 94, 11, 39, 20, 16, 5
 - i. Linear probing
 - ii. Double hashing where $R = 7$ and $m = 13$
4. a) Develop min heap for the given input and perform delete operation to insert it in a new array. Show min heap after each deletion. (6)
Input: 66, 33, 40, 20, 50, 88, 60, 11, 77, 30, 45, 65.

b) Devise pseudo code for single source shortest path (G, s). (4)
Input: $G = \{v, E\}$ and its vertex s.
Output: the length d_v of shortest path from s to v and its penultimate vertex p_v for every vertex v in G.

5. Find the minimum spanning tree for the below graph using both Prim's and Kruskal's algorithms. Which algorithm is more efficient in constructing the minimum spanning tree of a given graph Prim's algorithm or Kruskal's algorithm and why?

